

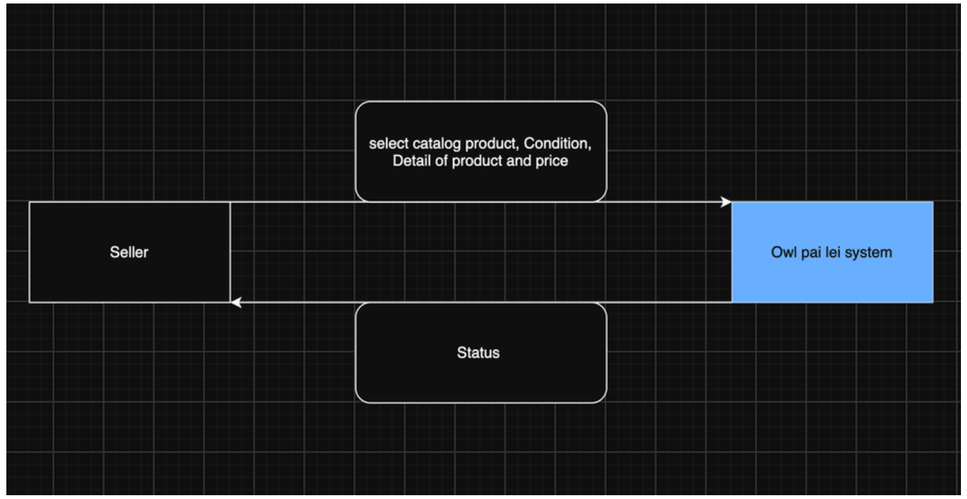
# OWL PAI LEI

# Context diagram:determine boundary

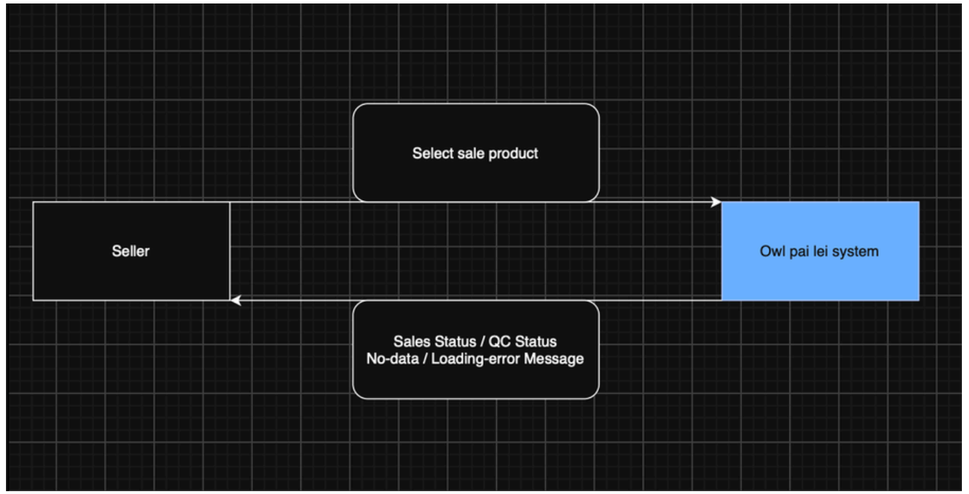
**US-S01: Offer a Product for Sale**
**As a seller, I want to select a product from Owl Pai Lei's catalog and submit its condition, defects, and asking price so that I can offer my item for sale.**

**US-S02: Track Sales and QC**
**As a seller, I want to view the sales and QC status of my items so that I can track their progress.**

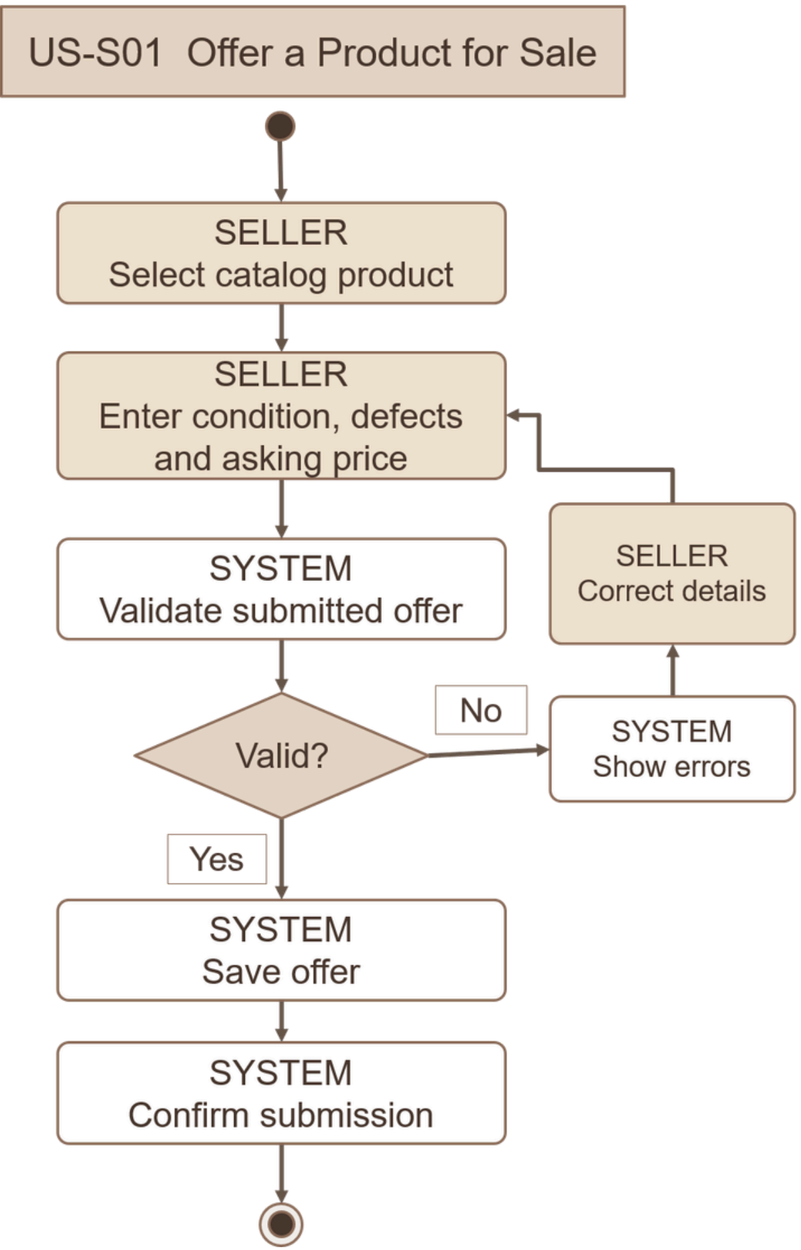
## US-S01: Offer a Product for Sale

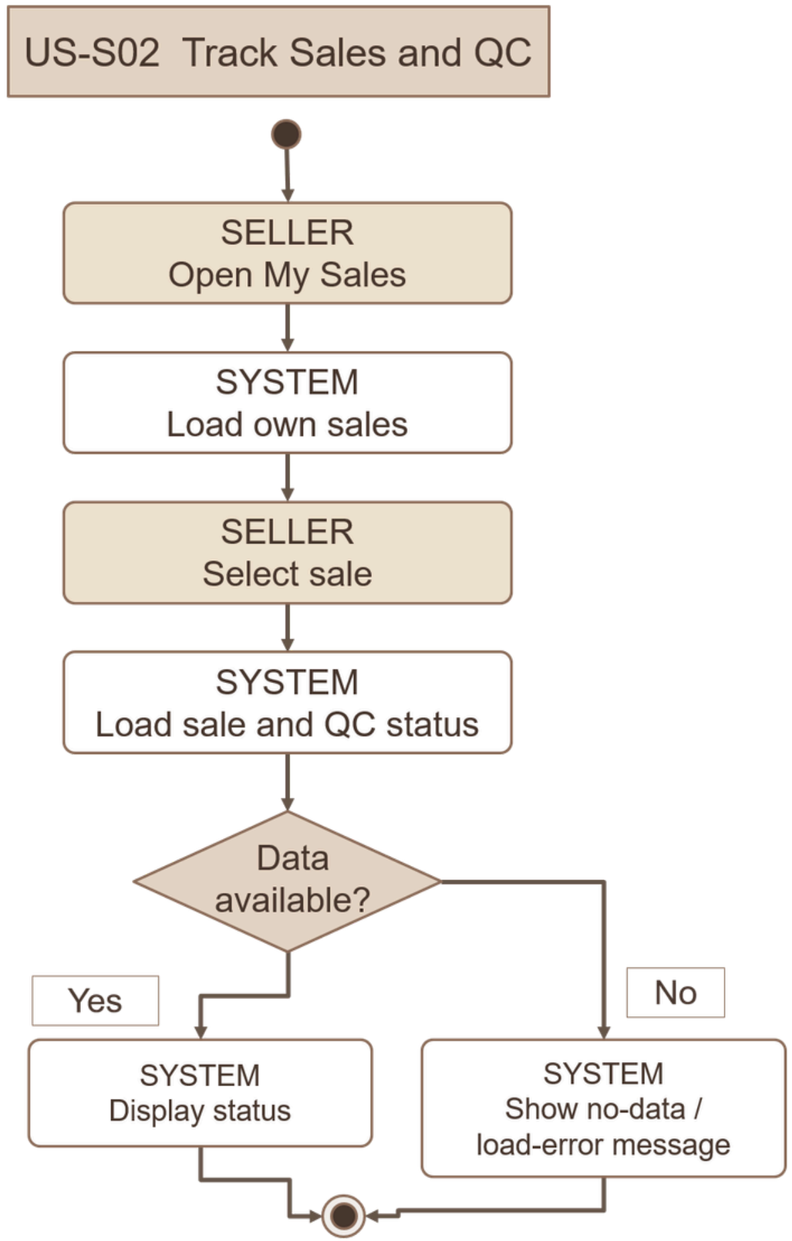



## US-S02: Track Sales and QC



# Activity Diagram Seller




**US-S01: Offer a Product for Sale**

The seller selects a product from Owl Pai Lei's catalog and submits its condition, defects, and asking price. The system validates the information. If valid, it saves the offer and confirms submission. Otherwise, it displays errors so the seller can correct and resubmit the details.

**US-S02: Track Sales and QC**

The seller opens My Sales and selects a sale. The system retrieves its sales and QC status. If the data is available, it displays the current status. Otherwise, it shows a no-data or loading-error message. Sellers can view QC results but cannot modify them.

# USE CASE

| | |
|---|---|
| **Use Case ID** | UC-S01 |
| **Name** | List Item for Sale |
| **Primary Actor** | Seller |
| **Goal** | Offer an item for sale from the catalog |
| **Trigger** | Seller selects Sell an Item |
| **Preconditions** | The product must have in catalog |
| **Success Postcondition** | The product save in catalog |
| **Failure Postcondition** | The product doesn’t save in catalog |

## Main senario

1. Seller opens Sell an Item.
2. System displays the product catalog.
3. Seller selects a product.
4. Seller chooses the condition
5. Seller enters the selling price.
6. Seller submits the listing.
7. System verifies that all required fields are filled.
8. System verifies that the price is valid.
9. System creates the listing.
10. System displays confirmation.

## Fail senario

- A1: Missing Information. At step 7, if the product, condition, or price is empty: System rejects the request. System displays “Please complete all required fields.”
- A2: Invalid Price. At step 8, if the price is zero, negative, or not a number: System rejects the request. System displays “Please enter a valid price.”
- A3: Product Not Found. At step 3, if the item is not in the catalog: System displays “This product is not available in the catalog.”

# USE CASE

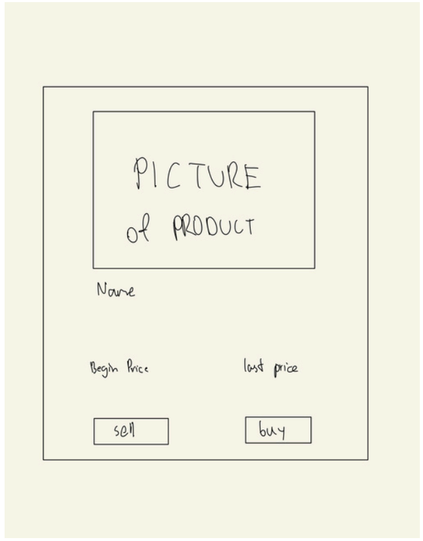
| | |
|---|---|
| **Use Case ID** | UC-S02 |
| **Name** | Track Sales and QC |
| **Primary Actor** | Seller |
| **Goal** | Track sales status and view Quality Control (QC) results |
| **Trigger** | click seller sell order, click product |
| **Preconditions** | Product must pusrchased |
| **Success Postcondition** | The status show |
| **Failure Postcondition** | - |

## Main senario

1. seller select catalog
2. seller select product
3. System show status

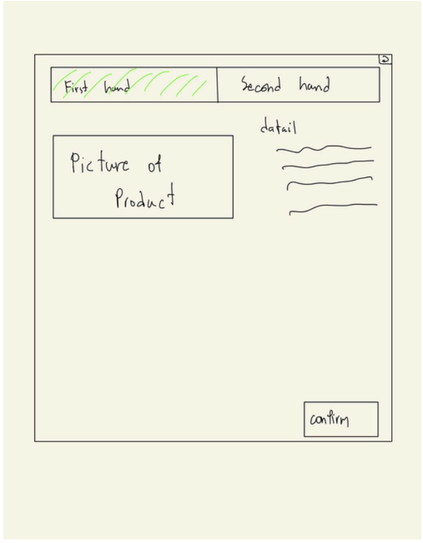
# UI WIRE FRAMES

choose product




click sell

choose first hand

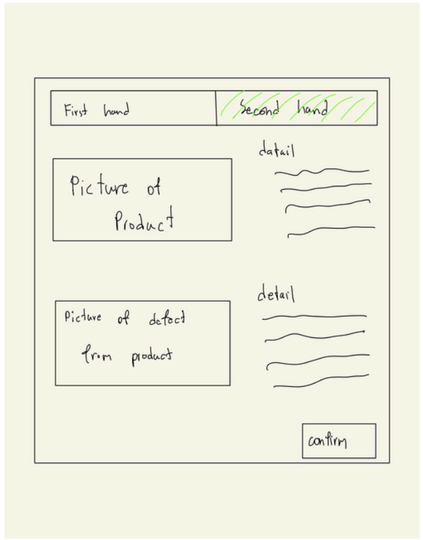



click confirm

result waiting



choose second hand

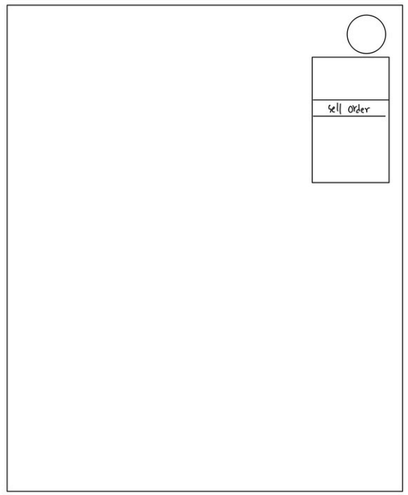



click confirm

result waiting

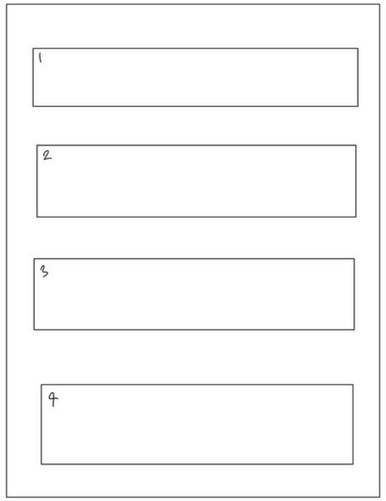


choose sell order



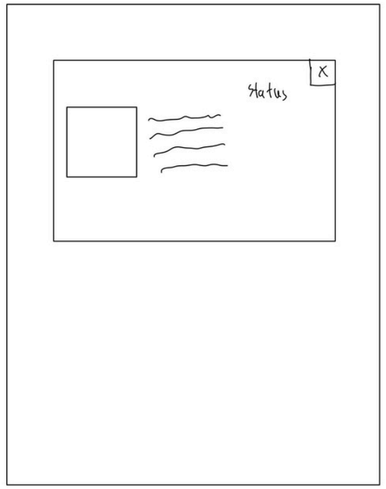

click sell order

choose produc

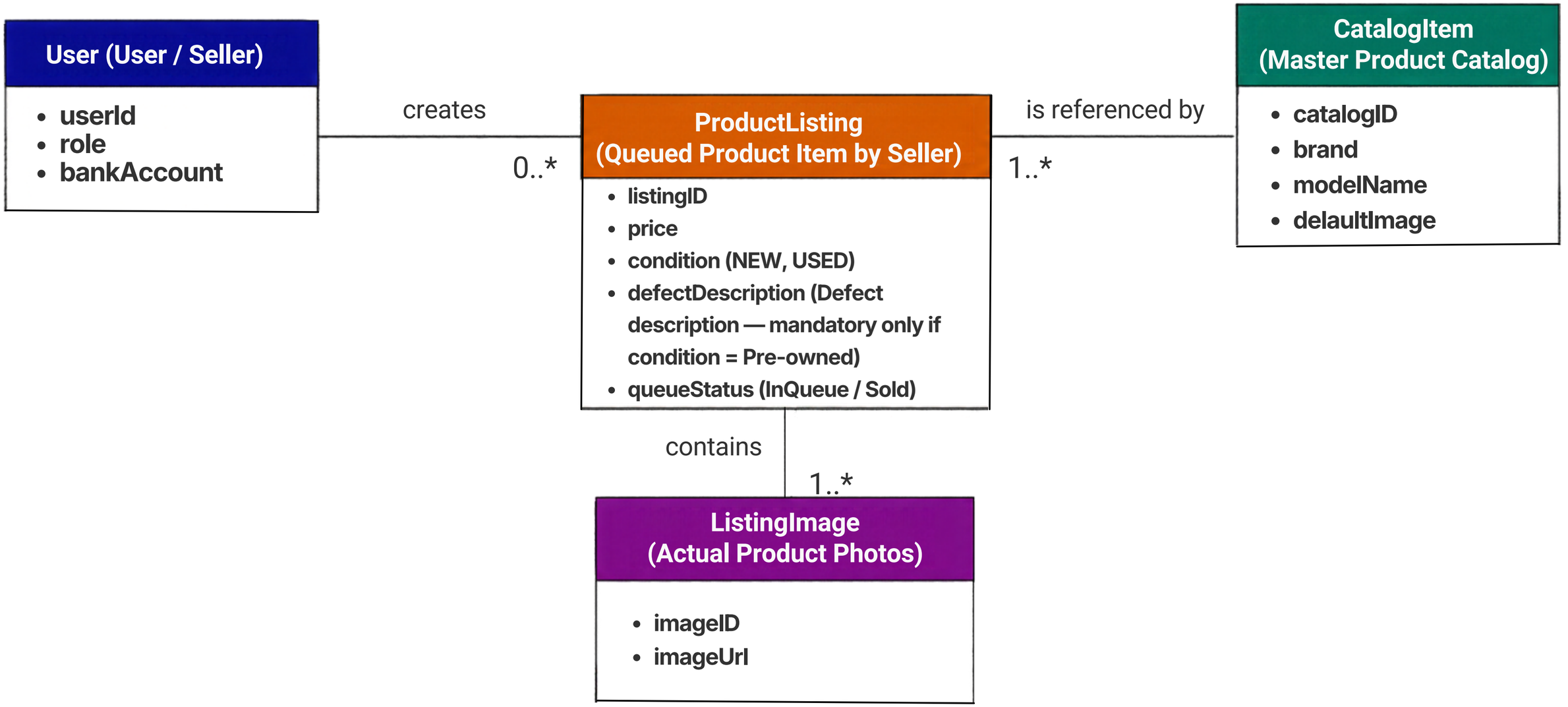



click product

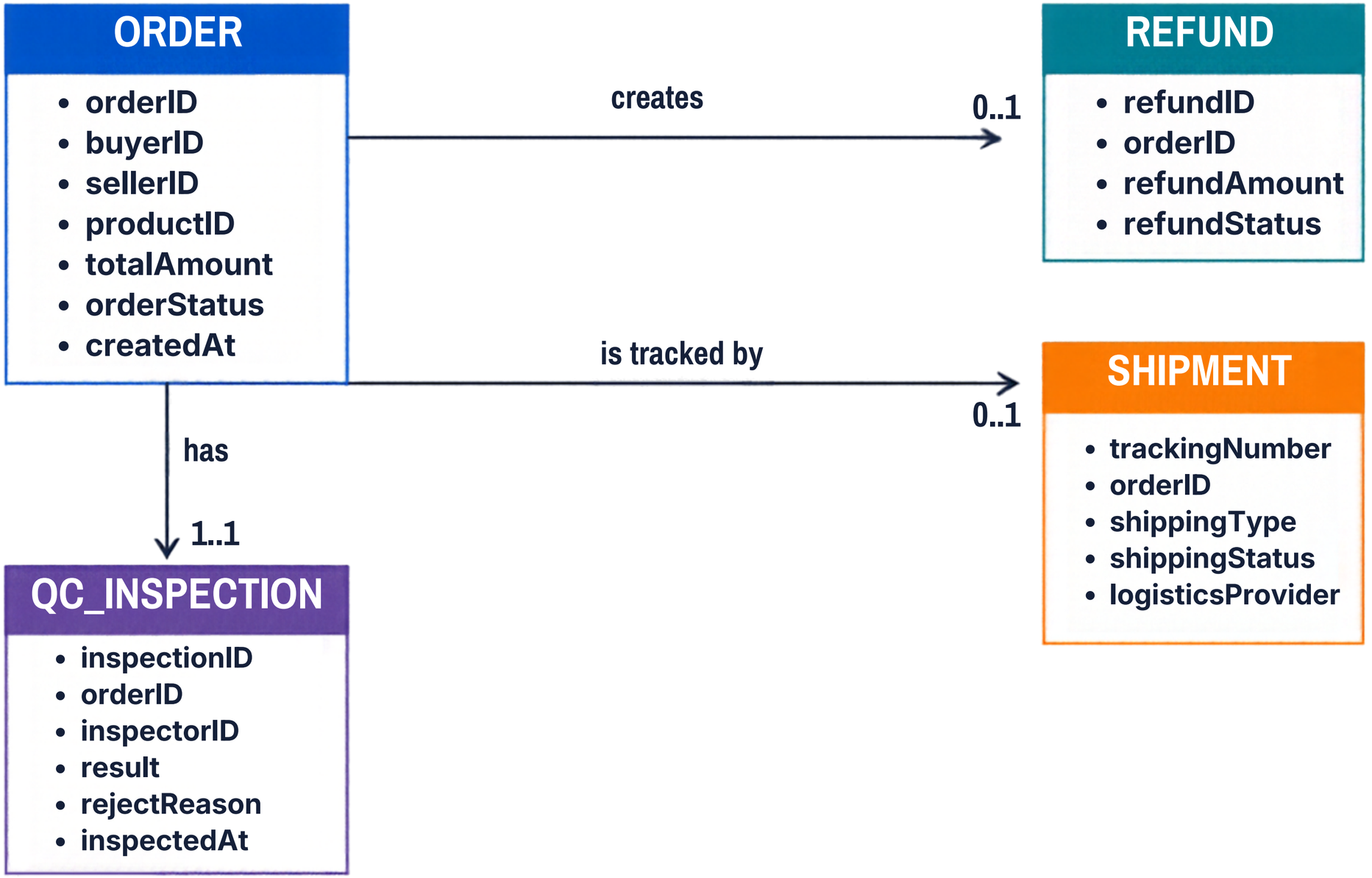
status show



# Domain Class Diagram :Seller Listing and Queue Management



# Domain Class Diagram : Order, Refund, Shipment and QC Inspection



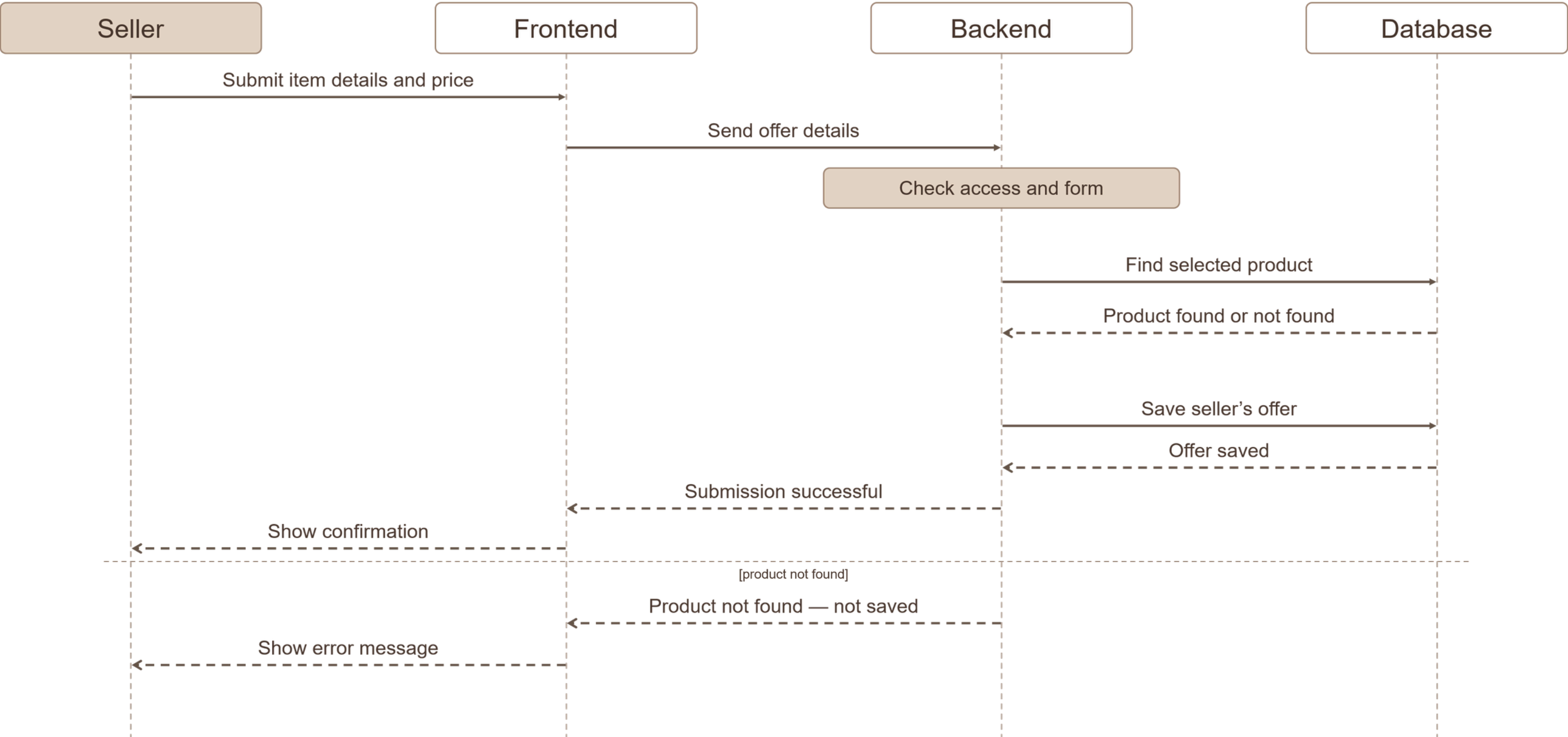
# Event / State Diagram – Seller

-A seller's item can be in Draft, Submitted, Availablefor Sale, Awaiting SellerShipment,  Awaiting QC.
-After inspection, it becomes Ready to Ship or QC Failed.
-Items that pass QC move to Shipped to Buyer, then Completed after the buyer confirms receipt and checks the item.



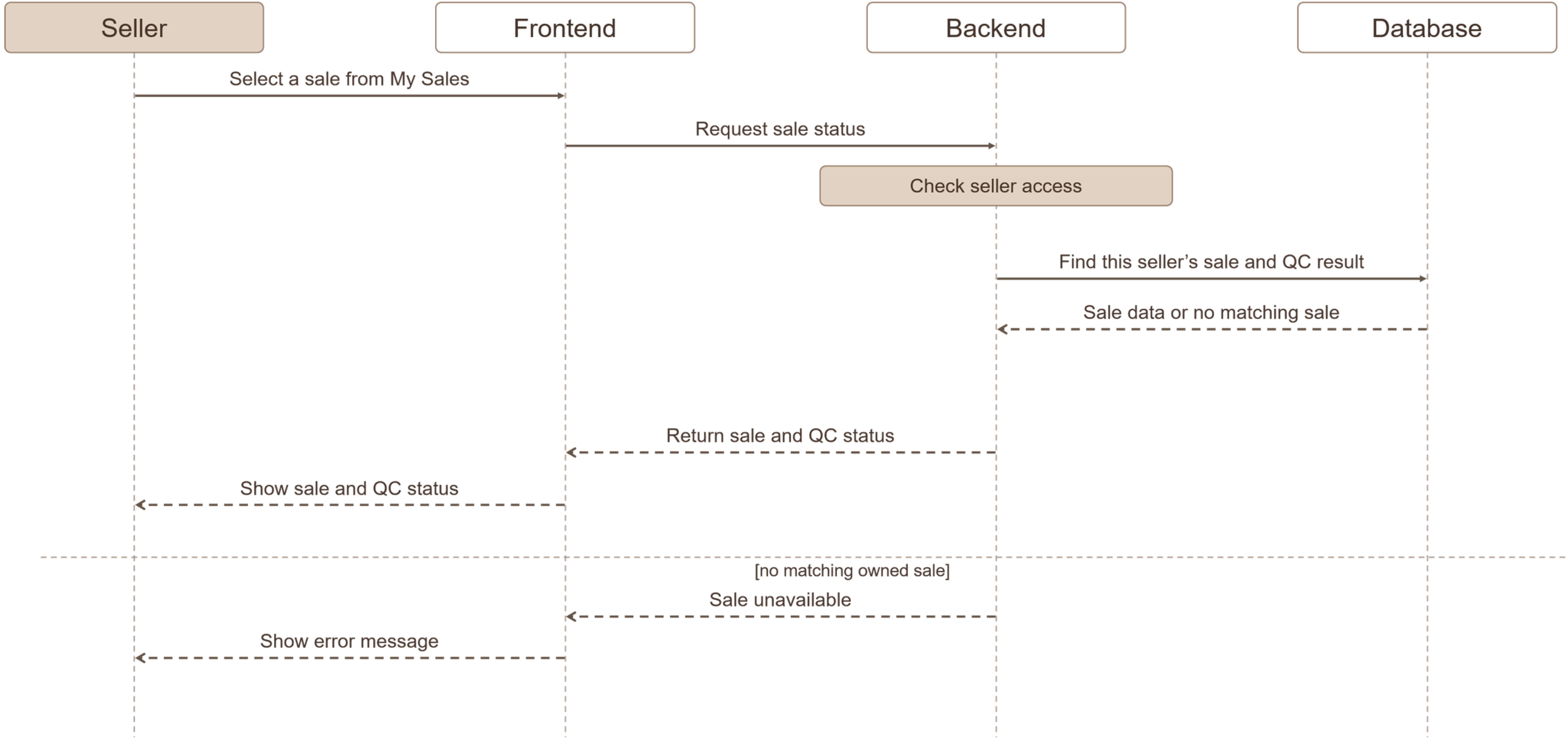
# Sequence Diagram — Seller

US-S01  Offer a Product for Sale

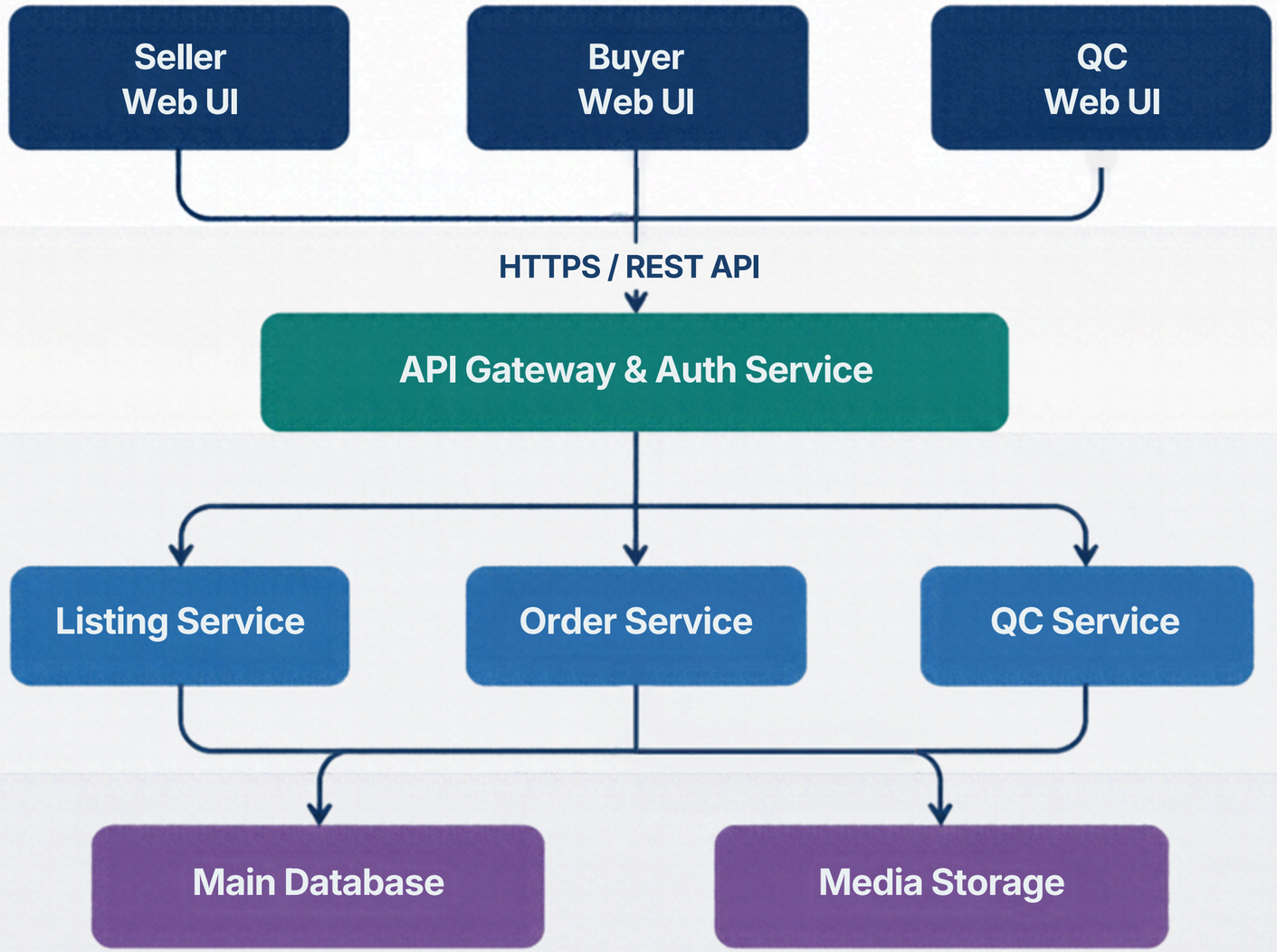


# Sequence Diagram — Seller

## US-S02  Track Sales and QC



# SYSTEM ARCHITECTURE DIAGRAM




**PRESENTATION LAYER**
User interfaces

**GATEWAY LAYER**
Request routing & authentication

**SERVICE LAYER**
Business logic & processing

**DATA LAYER**
Persistent storage

# 1. SELLER LISTING AND QUEUE MANAGEMENT

| API ENDPOINT | DESCRIPTION / FUNCTION | TARGET TABLE |
|---|---|---|
| GET /api/catalog | list available catalog items | CatalogItem |
| POST /api/listings | create a new product listing in queue | ProductListing |
| POST /api/listings/:id/images | upload product images | ListingImage |
| GET /api/listings/my-queue | list seller's queued items | ProductListing |

# ORDER, REFUND, SHIPPING AND QC INSPECTION

| API ENDPOINT | DESCRIPTION / FUNCTION | TARGET TABLE |
|---|---|---|
| POST /api/orders | create order and process payment | Order |
| POST /api/orders/:id/shipments | update shipment tracking status | Shipping |
| GET /api/qc/orders | list pending QC orders | Order |
| GET /api/qc/orders/:id | get order QC details | Order |
| POST /api/qc/orders/:id/inspections | submit QC result (auto-triggers refund if fail) | QC_Inspection, Refund |

# Business rule

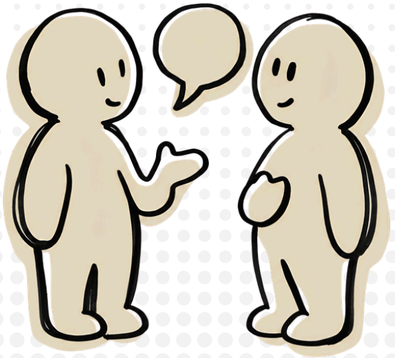
| Model | What is it like? | How it supports the rule |
|---|---|---|
| **1. Use Case** | **Instruction Manual** | States clearly in the steps: *"Before the buyer can proceed to payment, the cart must contain exactly 1 item. If there are 2 or more items, the system rejects the request."* |
| **2. Activity Diagram** | **Flowchart with Arrows** | Features a decision diamond asking: *"Is Cart Item Count = 1?"*<br><br>- If **Yes** → Arrow points to the Payment page.<br><br>- If **No** → Arrow points to an error message: *"Please purchase one item at a time."* |
| **3. State Diagram** | **Status Indicator** | The cart transitions from Selecting Items to Ready for Payment only if the guard condition [Item Count == 1] is true. Otherwise, it stays locked in the current state. |
| **4. Sequence Diagram** | **Backend Communication Chat** | The frontend sends a message: *"Customer A wants to checkout."*<br><br>The backend checks and replies: *"Denied! There are 2 items in the cart. Return an error message immediately."* |
| **5. Domain Class Diagram** | **Data Blueprint** | In the data structure, the multiplicity between the Order class and the OrderItem class is strictly set to **1** (meaning 1 Order can contain only 1 Item). |

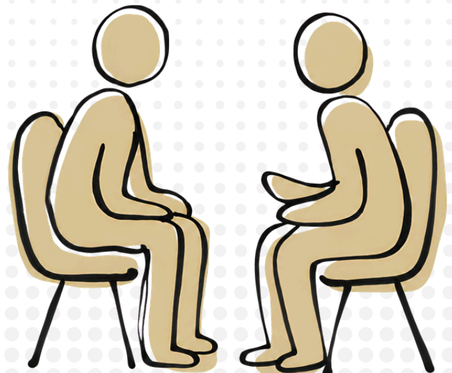
# UNRESOLVED DESIGN DECISION
# SELLER PRICE CHANGES

**Question: When should sellers stop being able to change their asking price?**

**Option 1:** Lock the price when an order is confirmed. This keeps the agreed price unchanged.

**Option 2:** Allow price changes until shipment. This gives sellers more flexibility but may cause payment disputes.





# Thank you

Owl Pai Lei